

# Yan Wang

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## EDUCATION

### Wuhan University

Wuhan, China

- B.S. in Biological Science; **GPA: 3.91/4.00; Major GPA: 3.97/4.00; Ranked #1** in the major Sep 2023- Jun 2027
- **Hongyi Honors Class:** University-Wide Selective Cohort For Advanced Research Training
- **Yale School of Medicine: Senior Thesis** Jul 2026 – May 2027

## SCHOLARLY OUTPUTS

### Manuscripts Under Review

**Wang, Y.,** Miller, L., Estrada, M., & Li, D. Mitochondrial dysfunction and  $\alpha$ -synuclein pathology in Parkinson's disease: A bidirectional pathogenic loop and therapeutic opportunities. *Frontiers in Molecular Neuroscience*. Review manuscript under review.

### Conference Abstracts

**Wang, Y.,** Zhang, J., Huang, X., Ruan, Y., & Yu, Y. Distributed INS-1 signaling through the candidate noncanonical receptor NTR-1 tunes the ASER – AIY axis to drive high-salt ethanol avoidance in *Caenorhabditis elegans*. Abstract accepted for presentation at *Neuroscience 2026, Society for Neuroscience*.

### Grants & Funding

Young Talent Sponsorship Program, Unitree Robotics. Project: “Multimodal monitoring of reversible hypometabolic states.” Application under review (anticipated decision Sept 2026).

## RESEARCH EXPERIENCE

### Adams Center for Parkinson's Disease Research, Yale School of Medicine

New Haven, CT

*Senior Thesis, Visiting Undergraduate Researcher | Mentor: Prof. Clemens Scherzer and Dunhui Li*

Jul 2026-Present

#### Project: Functional Dissection of the *PM20D1* Locus in Parkinson's Disease

- Investigate how *PM20D1* perturbation affects mitochondrial function and cellular redox homeostasis in human iPSC models using antisense oligonucleotides (ASOs) and gene-editing approaches.
- Characterize the subcellular localization of the lncRNA *PM20D1-AS1* and investigate its potential regulation of *PM20D1* expression, while exploring candidate molecular interaction partners.
- Independently develop RNA-seq analysis workflows to profile 22 annotated *PM20D1-AS1* transcripts, identifying preliminary evidence of increased relative abundance of specific isoforms in astrocytes and oligodendrocytes from patients with Parkinson's disease.

### Medical Research Institute, Wuhan University

Wuhan, China

*Undergraduate Researcher; Project I Lead | Joint lab of Prof. Yanxun Yu and Prof. Younghan Jin*

Jan 2025-Jul 2026

#### Project I: *INS-1-Dependent Neuropeptidergic Tuning of Sensory Valence in C. elegans*

- Independently elucidated the mechanism underlying the *ins-1*-dependent reversal of high-salt ethanol avoidance into attraction.
- Mapped distributed INS-1-releasing nodes in *C. elegans* by combining cell-specific *ins-1* deletion, rescue, and neuronal ablation, implicating ASER, AIY, AIZ, and AIB in ethanol avoidance.
- Demonstrated ASER-AIY circuit dysfunction using calcium imaging, showing a right-shifted ASER ethanol response range and weakened AIY negative calcium transients in *ins-1* mutants.
- Revealed ASER-derived INS-1 as a major regulator of ASER sensory tuning. Integrated ethanol dose-dependent behavioral assays with physical modeling of diffusion, supporting a quantitative model in which INS-1 tunes ethanol sensitivity rather than causing simple sensory loss.
- Proposed NTR-1 as a candidate noncanonical INS-1 receptor by integrating pathway mutant analysis, *ntr-1* phenotyping, molecular docking and MD simulations.
- Spearheaded a circuit-level model in which distributed INS-1 release, potentially mediated by NTR-1, calibrates ASER sensory gain and downstream AIY output to code ethanol valence under high-salts.

#### Project II: Performance Optimization of the POST-IT Proximity Tagging System

- Optimized POST-IT, a Pupylation-based non-diffusive proximity-labeling platform for live-cell-compatible small-molecule target identification.

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- Engineered PafA ligase to reduce the major kinetic and substrate constraints of the original POST-IT system, including long labeling time and high SBP-Pup (E) requirement.
- Built a yeast-surface-displayed Pep1-MATH weak-interaction selection system to couple PafA catalytic activity with FACS-based enrichment of improved variants.
- Executed iterative directed evolution of a  $\sim 10^6$ -scale PafA mutant library constructed via error-prone PCR, integrating FACS enrichment with NGS-guided mutation tracking under progressively stringent labeling conditions to isolate high-performing variants.
- Improved labeling from 18h/40 $\mu$ M to 2h/10 $\mu$ M, increasing threshold-positive cells from 0.5% to 27.2% and achieving >40-fold apparent performance enhancement.

**College of Life Science, Wuhan University**

Wuhan, China

*Wet Lab Leader; Project Prototype Designer, 2025 iGEM Gold Medal | Advisor: Assoc. Prof. Lei Bai*

Nov 2024-Oct 2025

Project: LOGIC -Lasting Optimized Genetic Input Calculator

- Originated and designed LOGIC, utilizing quorum-sensing-mediated intercellular communication as programmable spatial “wiring” for colony-level bio-computation.
- Designed a diffusion-based biocomputing architecture in which orthogonal AHL signaling channels, engineered sender-receiver colonies, and spatial concentration gradients are integrated to implement modular genetic logic beyond single-cell circuit constraints.
- Built and characterized high-pass and band-pass response modules to encode Boolean operations, construct AND, OR, XOR, multi-input gates, and cascaded logic operations on solid media.
- Developed a serial biological adder framework by coordinating input, output, and carry signals, advancing the system from isolated logic gates toward executable biological arithmetic.
- Integrated optogenetic refresh modules, including split-VVD AiiA for light-controlled AHL degradation and LOVdeg-tagged output proteins for signal reset.

## TECHNICALSKILLS

**Model Organisms & Cell Culture:** *E. coli*, *S. cerevisiae*, *C. elegans* manipulation; human iPSC culture and maintenance

**Molecular & Biochemical Methods:** Molecular cloning; RNA techniques (e.g., qPCR, RNAscope, RNA pull-down); protein expression and purification; Western blot; co-immunoprecipitation; FACS screening; general biochemical assays

**Neuroscience & Imaging Techniques:** *C. elegans* — neuronal calcium imaging, optogenetic manipulation, microinjection, genetic crossing, and behavioral phenotyping; confocal microscopy

**Computational & Structural Biology:** snRNA-seq data analysis; customized Python/R analytical pipelines; molecular docking; molecular dynamics (MD) simulations

## HONORS & AWARDS

**Westlake University Rising Star Award** (¥30,000), top 9 out of 300 candidates Aug 2026

**Liu Daoyu Creative Learning Scholarship** (¥8,000), awarded to top 24 university-wide students Jan 2026

**Gold Medal**, International Genetically Engineered Machine Competition (iGEM) Nov 2025

**Multiple Gold Prizes**, Regular Track, Synthetic Cell Track, Protein Design Track, Bacterial Game Track | **SynBio Challenges** (International Synthetic Biology Competition) 2025–2026

**Hongyi Honors College Annual Top Ten Science Advances Award** Oct 2025

Outstanding Student Scholarship (top 5% cohort), Outstanding Student & Outstanding Student Leader 2024–2025

First-Class Outstanding Freshman Scholarship 2023

## ACADEMIC & SCHOLARLY TRAINING

Participant, Neuroscience & Cognitive Science Summer Training Program, **Peking University** Jul 2026

Participant, “Frontiers of Life Sciences” International Summer School, **Westlake University** Jul–Aug 2025

Delegate, National “Top-Tier Talent 2.0” Biology Forum, **Tsinghua University** May 2025